

Sustainable Heat & Power Generation

Built Environment

May, 2026

Sustainable Heat & Power Generation (A18801)

Short Title: Sustainable Heat & Power
Generation
Faculty Science and Computing
Department Built Environment
Cluster Engineering
Author RGREENE
Credits 5

Level: Intermediate

Description of Module / Aims

The aim of this module is to provide a specialised knowledge of sustainable heat and power generation.

Programmes

BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI)	2 / 3 / M
BEng (Hons) in Sustainable Energy Engineering (WD_ESERG_B)	/ /

Indicative Content

- 1. Wind, hydro and solar power
- 2. CHP and trigeneration
- 3. Nuclear power
- 4. Energy storage technology and fuel cells
- 5. Waste heat recovery and utilisation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. On successful completion of this module, students will be capable of explaining the principles of heat and power generation.
2. Analyse the characteristics and performance of alternative energy systems.
3. Apply standard calculation techniques to solve problems involving energy supply and demand.
4. Demonstrate an understanding of energy storage technology.
5. Recognise and understand the comparative cost benefits of different low and zero carbon technologies.

Learning and Teaching Methods

- 1. Lectures
- 2. Presentations
- 3. Interactive Practicals
- 4. Reading
- 5. Exercises
- 6. Discussion

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	100%	1, 2, 3, 4, 5

Assessment Criteria

Learning Modes

Learning Mode	F/T Hours	P/T Hours
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Essential Material

- 1. Godfrey Boyle, 2004, Renewable energy power for a sustainable future, 2nd Edition, Oxford University Press
- 1. Lecture
- 2. Aldo V. Da Rosa, 2005, Fundamentals of Renewable Energy Processes, Elsevier
- 2. Independent learning
- 3. Godfrey Boyle, Energy Systems & Sustainability, Oxford University Press
- 4. CIBSE guides and technical notes
- 5. BSRIA guides and technical notes
- Learning modes
- Reading materials